

PRACTICE QUESTIONS – TOPIC : SET

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*set pratice question.py - C:/Python314/set pratice question.py (3.14.3)*
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#1)what is mean by "SET" in the python ?
print("set")
print("----")

print("set is a collection of unique elements in python ,its unordered,unindexed,
      "it's not allow duplicate values, we can edit the elements using set method")

#2)what happens if duplicate values are added in a set ? explain with an example.
print("set doesn't allow duplicate values while creating a set the duplicate value")
print("-----")
a={2,3,4,4,6,6,8,8,9}
print(a)

#3)is a set is mutable or immutable ? explain with an example.
print(" set is mutable because we can add or remove the elements using set methods")
print("-----")
a={2,3,4,6,9}
a.remove(4) #remove- method of set # proves set is mutable
print(a)

#4)what is the difference between list and set in python?
print("list is ordered,indexed but set is unordered ,unindexed" ,
      "list supports duplicate elements but set remove duplicate automatically "
      "both are mutable and heterogeneous (store multiple datatypes)")
#5)what is the difference between tuple and set in python?
print("tuple is ordered,indexedbut set is unordered",
      "unindexed and tuple allows duplicate elments",
      " but set remove duplicate elements automatically and tuple is immutable ,set")
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===== RESTART: C:/Python314/set prattice question.py =====

1)set

set is a collection of unique elements in python ,its unordered,unindexed,mut
able,doesn't supports indexing it's not allow duplicate values, we can edit t
he elements using set methods

2)set doesn't allow duplicate values while creating a set the duplicate value
s willnot store

{2, 3, 4, 6, 8, 9}

3)set is mutable because we can add or remove the elements using set methods
like add(),remove()

{2, 3, 6, 9}

4)list is ordered,indexed but set is unordered ,unindexed list supports dupli
cate elements but set remove duplicate automatically both are mutable and het
erogeneous (store multiple datatypes

5)tuple is ordered,indexedbut set is unordered unindexed and tuple allows dup
licate elments but set remove duplicate elements automatically and tuple is
immutable ,set is mutable

{65, 33, 98, 67, 76}

{'honda', 'toyota', 'BMW'}

>>>

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print("5)tuple is ordered,indexedbut set is unordered",
      "unindexed and tuple allows duplicate elments",
      " but set remove duplicate elements automatically and tuple is immutable ,se

#6) how do you add elements to a set ?
#create a set include the values
#use (add) dotted function of set to add the value in set
#use built in function (print)
print("6)how do you add elements to a set ?")
print("-----")

student_marks ={76,98,33,65}
student_marks.add(67)
print(student_marks)

#7)how do you remove elements from set ? explain with example.
print("7) how do you remove elements from set ? explain with example.")
print("-----")

car = { "toyota" , "BMW" , "audi" , "honda" }
car.remove("audi") #used (remove) dotted function of set
print(car)
```

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===== RESTART: C:/Python314/set prattice question.py =====
6)how do you add elements to a set ?
-----
{65, 33, 98, 67, 76}
7) how do you remove elements from set ? explain with example.
-----
{'toyota', 'honda', 'BMW'}
>>>
===== RESTART: C:/Python314/set prattice question.py =====
using discard method
-----
{'maths', 'tamil', 'social', 'science'}
using remove method
-----
Traceback (most recent call last):
  File "C:/Python314/set prattice question.py", line 63, in <module>
    books.remove("english")
KeyError: 'english'

```

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#what is the difference between remove() and discard()method in set?
#the difference between remove and discard are:
#if we use remove method - it shows error if the given specific element is not in
#if we use discard method - it dosen't show error if the given element is not in s
print("using discard method ")
print("-----")
books = { "maths" , "science" , "social" , "tamil"}
books.discard("english")
print(books)

print("using remove method")
print("-----")
books = { "maths" , "science" , "social" , "tamil"}
books.remove("english")
print(books)

#8) can we access set elements using index ? why?
print("8)we can't access the set because set is unordered and unindexed collection
      "only we use index in list ,tuple , string because they all ordered,indexed"
a={4,6,7,8}
a.index(2)
print(a) #it give error

#9)what is the difference between pop() in list and pop() in set?
print("9)pop() in the list removes the last element in the list",
      "opo() in the set removes random element in set bcoz set is unordered, uninc
print('using pop() in list')
print("-----")
name=["priya" , "kumara" ,"padma" , "pavithra" ]
name.pop() #removed last element
print(name)
|
print("using pop() in set")
print("-----")
name={"priya" , "kumara" ,"padma" , "pavithra" }
name.pop() #removed random element
print(name)

```

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===== RESTART: C:/Python314/set pratice question.py =====
6)how do you add elements to a set ?
-----
{65, 33, 98, 67, 76}
7) how do you remove elements from set ? explain with example.
-----
{'toyota', 'honda', 'BMW'}
>>>

===== RESTART: C:/Python314/set pratice question.py =====
using discard method
-----
{'maths', 'tamil', 'social', 'science'}
using remove method
-----
Traceback (most recent call last):
  File "C:/Python314/set pratice question.py", line 63, in <module>
    books.remove("english")
KeyError: 'english'
>>>

===== RESTART: C:/Python314/set pratice question.py =====
8)we can't access the set because set is unordered and unindexed collectionon
ly we use index in list ,tuple , string because they all ordered,indexed
Traceback (most recent call last):
  File "C:/Python314/set pratice question.py", line 70, in <module>
    a.index(2)
AttributeError: 'set' object has no attribute 'index'
>>>

===== RESTART: C:/Python314/set pratice question.py =====
9)pop() in the list removes the last element in the list opo() in the set rem
oves random element in set bcoz set is unordered, unindexed
using pop() in list
-----
['priya', 'kumara', 'padma']
using pop() in set
-----
{'padma', 'kumara', 'pavithra'}

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#10) explain union operation in set with an example.
#union - combine all the elements from two set and remove the repeated element
print("10)explain union operation in set with an example.")
print(" -----")
float_1 = { 3.4 , 4.9 , 1.2 ,8.2 }
float_2 = {1.2 , 4.9 , 5.3 ,8.2 ,7.8}
float_value = float_1.union(float_2)
print(float_value)

#11)explain intersection operation in set with an example.
#intersection - it return the common elements from two set
print("11)explain intersection operation in set with an example.")
print(" -----")
float_1 = { 3.4 , 4.9 , 1.2 ,8.2 }
float_2 = {1.2 , 4.9 , 5.3 ,8.2 ,7.8}
float_value = float_1.intersection(float_2)
print(float_value)

#12)explain the difference operation in set with an example
# it return the elements from first set which is not in second set
print("12)explain the difference operation in set with an example")
print(" -----")
dark_color = {"red" , "darkbrown" , "green" , "black"}
light_color = {"skyblue" , "yellow" , "pink" , "black"}
color = dark_color.difference(light_color)
print(color)

#13) how do you find common elements between two sets?
#using intersection method we can find common elements
#it return the common element from two sets
print("13) how do you find common elements between two sets?")
print(" -----")
fruits_1 = { "orange" , "apple" , "pineapple" , "watermelon"}
fruits_2 = { "strawberry" , "jackfruit" , "mango" , "pineapple" , "apple"}
fruits = fruits_1.intersection(fruits_2)
print(fruits)
```

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using pop() in list
-----
['priya', 'kumara', 'padma']
using pop() in set
-----
{'padma', 'kumara', 'pavithra'}
10)explain union operation in set with an example.
-----
{1.2, 3.4, 4.9, 5.3, 7.8, 8.2}
11)explain intersection operation in set with an example.
-----
{8.2, 1.2, 4.9}
12)explain the difference operation in set with an example
-----
{'red', 'green', 'darkbrown'}
13) how do you find common elements between two sets?
-----
{'pineapple', 'apple'}
```

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#14) how can you remove duplicates from a list using a set? explain with an example
# we can remove duplicate from a list by converting
#the list into set bcoz set doesn't support duplicate
print("14) how can you remove duplicates from a list using a set? explain with an
print("-----")
a = [23,56,78,98,56,23,98,45,36]
b = (list(set(a))) #set removes duplicate automatically but the result is unordered
print(b)
#if we want the result should be in order after removing duplicate there are 2 ways
#dic.fromkeys()
#using append method

print("using dic.fromkeys() method from dictionary")
print("-----")
a = [23,56,78,98,56,23,98,45,36]
b = list(dict.fromkeys(a)) #in dictionary the key it doesn't allow duplicate values
print(b)

print("using append method from list ")
print("-----")
a = [23,56,78,98,56,23,98,45,36]
b=[]
for num in a:
    if num not in b :
        b.append(num)

print(b)

#15) what is the difference between add() and update() method in a set? explain with
#add() - it add the "single" element in set
#update - it add "multiple" element in set
print("15) what is the difference between add() and update() method in a set? explain
print("-----")
print("using add() method ")
print("-----")
float_num = { 3.4 , 4.9 , 1.2 ,8.2 }
float_num.add(1.2)
print(float_num)
```

>>>

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===== RESTART: C:/Python314/set prattice question.py =====
14) how can you remove duplicates from a list using a set? explain with an example
-----
[98, 36, 45, 78, 23, 56]
using dic.fromkeys() method from dictionary
-----
[23, 56, 78, 98, 45, 36]
using append method from list
-----
[23, 56, 78, 98, 45, 36]
15) what is the difference between add() and update() method in a set? explain with example
-----
using add() method
-----
{8.2, 1.2, 3.4, 4.9}
using update() method
-----
{1.2, 3.4, 4.9, 3.5, 8.2, 9.4}
```

